

Integration Services

▼ About Integration Services

The integration use cases enable services to be exposed and consumed.

The integration services cover both point-to-point integrations and integrations using event-driven architectures. ABAP Cloud enables integration between SAP cloud products, SAP BTP services, on-premise landscapes, customer extensions, and external services. The exposure of services to other external systems is referred to as inbound communication. The consumption of services from external systems is referred to as outbound communication. Integration services in ABAP Cloud are used for process or data integration. Process integration structures communication at the app level, along with a predefined business process, for example, order-to-cash. Data integration addresses the data exchange between two or more communication partners without a relation to a business process. One of the main use cases for data integration is cross-system analytics. Both integration use cases rely on the same architecture, but use different protocols to achieve the respective integration

▼ Data and Process Integration Comparison

The following table gives you an overview of both integration service scenarios, outlining the main differences and main use cases.

Comparing Process and Data Integration

Dimensions	Process Integration	Data Integration
Scenario Scope	- Integrates several apps, along a predefined business process - Domain-specific logic is taken into account for all operations. (Potentially) Bidirectional communication between communication partners	- Integrates several communication partners without relation to any specific business process. - Domain-specific logic is bypassed (only READ logic is considered). - One-directional communication between communication partners
Domain-Specific Implementation	Write/Updates always through domain-specific implementation layer (logic is considered).	Write/Updates always directly on database layer.
Data Representation	Exchanged data is converted to an external format on provider side (for example ISO Currencies).	- Exchanged data keeps source types (for example ABAP dates) to avoid any loss of data or semantics because of conversions. - Only minimal conversions are applied, which can only be processed on provider-side (for example decimal shift).

Dimensions	Process Integration	Data Integration
Protocol	• OData • SOAP • Business Events • Remote Function Call	• SQL

▼ Protocols for Process Integration

Available Integration Protocols

The goal of process integration is to facilitate communication between multiple apps within a predetermined business process. ABAP Cloud offers many different protocols for integration, for these reasons:

- The communication partner (SAP products or 3rd party) usually exposes a functionality only through a dedicated protocol.
- Different use cases for integrations require different protocols, for example for asynchronous integrations for using event-driven architectures.

The following table gives you an overview of the available protocols:

Protocols for Point-To-Point Integration

Protocol Name	Description	When To Use	Features	Constraints	Available for Inbound Communciation	Available for Outbound Communciation
OData	Open Data Protocol (OData) is an open protocol that allows the creation and consumption of queryable and interoperable REST APIs in a simple and standard way.	• Synchronous process integration with modern ABAP systems	• Standard-based protocol and data format • Batch requests • System query options	• Limited asynchronous options	Yes	Yes

Protocol Name	Description	When To Use	Features	Constraints	Available for Inbound Communciation	Available for Outbound Communciation
SOAP (Simple Object Access Protocol):	SOAP services are Web services that enable data exchange based on a standardized XML format through the Simple Object Access Protocol.	Process integration with existing APIs	- Standard-based protocol and data format - Synchronous and asynchronous options - Support for exactly-once-in-order pattern - AIF interfaces for monitoring of asynchronous processing	- Supports only the generation of service consumers from external metadata. - No native integration with RAP	No	Yes
Remote Function Call (RFC)	RFC is a proprietary SAP interface for communication between ABAP systems RFC Connectors 	- Synchronous process integration with old ABAP systems or for high-performance requirements - Data replication with old ABAP systems	- Client libraries available - Compatibility with old ABAP releases - High performance	- No support for exactly-once-in-order patterns (bgRFC) in ABAP Cloud - No integration with RAP - Only integration with ABAP or RFC Connectors 	Yes	Yes
SQL	The ABAP SQL Service can be used to expose CDS view entities to external, ODBC-based SQL clients.	Synchronous data integration interface for data federation and replication	- Client libraries available - Consistency and performance for large payloads	- No standardized data serialization format - Only consumption through ODBC Driver	Yes	No

Protocol Name	Description	When To Use	Features	Constraints	Available for Inbound Communciation	Available for Outbound Communciation
HTTP	You can use the full flexibility of HTTP protocol by using the HTTP client library to send a request to any HTTP endpoint, or provide your own HTTP service for other integration partners.	Full flexibility of HTTP protocol	Flexibility to adapt implementation to specific use case requirements	- No integration with RAP - No standardized data serialization format	Yes	Yes

Event-Based Integration

Name	Description	When To Use	Features	Constraints	Available for Event Provider	Available for Event Consumer
Business Events	Event-driven architecture enables asynchronous communication between an event provider and consumer in use cases where no direct response from the consumer is required.	Asynchronous process integration, using event-driven architectures	- Asynchronous integration - Standard-based data format - Customized payload, with derived events based on SAP delivered events - AIF interfaces for monitoring of asynchronous processing	- Only for integration through event-driven architectures - No support for in-order processing	Yes	Yes

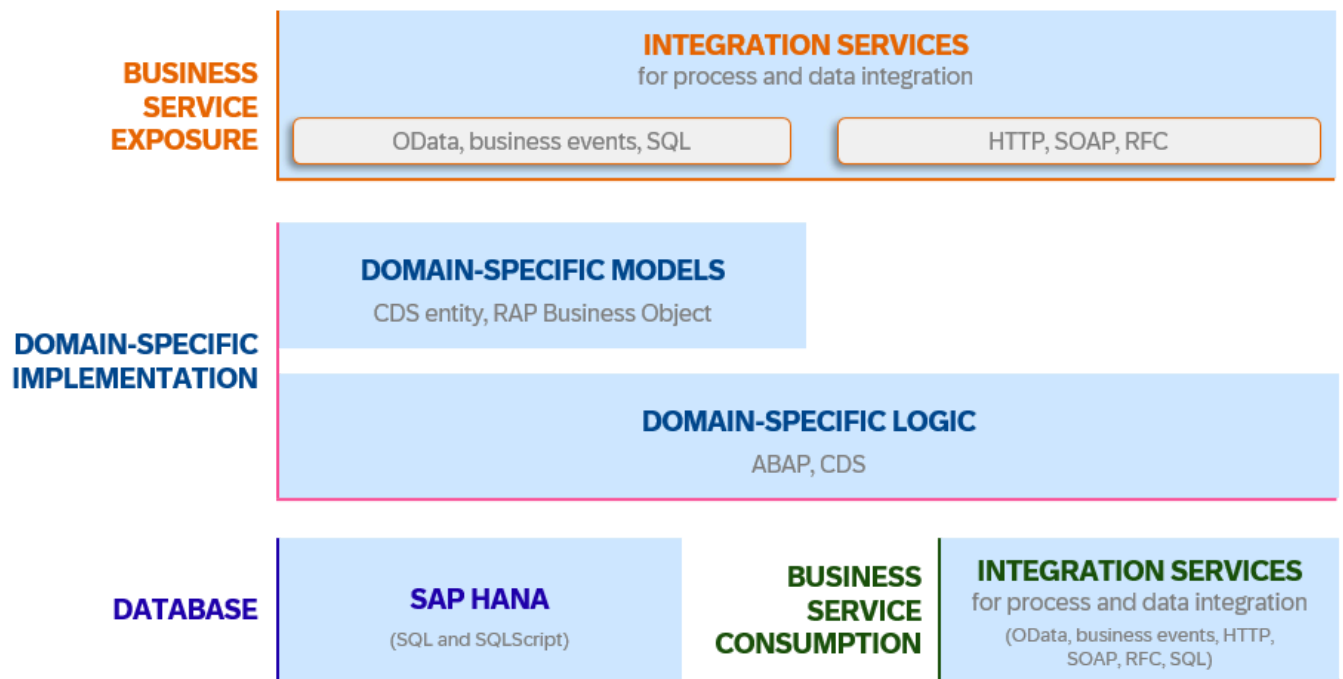
Architecture

The development model for exposing integration services depends on the respective protocol.

- Integration Service Exposure: OData, business events, and SQL can be created using the architecture and development objects of the RAP programming model. The domain-specific logic of SOAP, HTTP, and RFC services is implemented with ABAP.

- **Integration Service Consumption:** The consumption of services and events from other systems is simplified by using metadata to generate an Event Consumption Model and a Service Consumption Model. Service Consumption Models are available for OData, SOAP, RFC, and business events. For HTTP and RFC, you can define the consumption requests directly in ABAP.

With communication scenario as a development object, you can bundle inbound and outbound integration services to simplify the setup process for the administrator. The communication scenario defines the protocols that are used for the integration.



Develop or Consume Integration Services

▼ Recommended Readings

- [Integration Use Cases - Initial Considerations](#)
- [Available Integration Protocols](#)
- [Conceptual Background: Access Management \(ABAP Cloud Guide\)](#)

Get Started

You can use this [tutorial](#) to learn how to expose an ABAP CDS view with a communication scenario. This introduces the basic flow of an integration service and the development objects required to expose an OData service in an inbound communication scenario. This [chapter](#) of the ABAP Cloud guide introduces the design-time development objects necessary to establish cross-system communication - this is mainly relevant for the administrator persona, but also helps you as developer to get the big picture of the communication management design-time.

For event-based communication, you can watch [Events for ABAP Cloud-based Applications](#) that introduces the topic. You can also read this [chapter](#) of the ABAP RESTful Application programming model guide to get to know the conceptual background about business events in ABAP Cloud. You can implement your own event using this [example](#) in the same documentation.

Develop

For **data integration scenarios**, you can expose CDS data models as SQL services. This [chapter](#) gives you an overview of the relevant background for data integration with SQL services. You can use this [example](#) in the documentation to implement and expose your own SQL service. You can also use the tutorial [Using ODBC Driver for SQL service](#) to implement your own example.

For **process integration**, you can implement a Remote Function Call to get data from an on-premise system with the tutorial [Get Data from an On-Premise System Using a Remote Function Call](#). This gives you a good overview of the end-to-end process.

To develop a more advanced **business event use case**, you can use the tutorial [How to Create RAP Business Events in SAP BTP ABAP Environment](#). You can also implement derived events: A derived event is defined with reference to an already implemented event and allows the definition of a payload that deviates from the one of the referenced event. This [chapter](#) in the RAP documentation introduces the conceptual background and implements an example.

Consume

For consumption in **data integration use cases**, you can refer to the chapters [Consumption of the SQL Service for Data Federation](#) and [Consumption of the SQL Service for Data Replication](#) in the documentation.

The chapter [Consume External Integration Services \(Outbound Communication\)](#) in the ABAP Cloud guide gives you an overview of the consumption use cases for **data and process integration** for all protocols and links to more detailed information for how to implement the consumption use case.

The chapter [Consume Business Events](#) gives you a short overview of the consumption for business events and how to implement it. You can also use this [tutorial](#) to implement an event consumption.

Monitor and Configure

Integration services require communication management to enable the integration of different solutions and enable data exchange between the integrated solutions and processes. This [chapter](#) of the ABAP Cloud guide gives you an overview of communication management for the public cloud products that offer ABAP Cloud.

Deploy

Development objects created for integration services rely on the respective lifecycle management of the respective products. For more information, see [Lifecycle Management \(ABAP Cloud guide\)](#).

More Information and Support

- [SAP community page](#) for ABAP connectivity: Introduces ABAP connectivity technologies for remote on-premise and cloud-based SAP- and non-SAP systems. Here you can ask the SAP community for support in case you have questions.
- [ABAP resource library for integration services](#): Find an overview of all available resources for developing integration services.